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Solution of the Schrödinger Wave Equation in a Space-Time Continuum by a Finite Element Formulation

Eine finite Element-Methode zur Lösung der komplexen dreidimensionalen zeitabhängigen Schrödinger-Gleichung wird formuliert. Die bestimmenden finiten Elementgleichungen sind in jedem orthogonalen, ebenen oder gekrümmten Koordinatensystem gültig. Ausgehend von den allgemeinen Elementgleichungen werden zwei spezielle finite Elemente entwickelt, deren Elementmatrizen alle in expliziter Form angegeben sind.

A finite element formulation for the solution of the complex three dimensional time dependent Schrödinger equation is developed. The governing finite element equations are valid in any orthogonal flat or curvilinear coordinate system. From the general element expressions two specific finite elements are developed and all element matrices are explicitly given.

Для решения комплексного трёхмерного, зависящего от времени, уравнения Шрёдингера формулируется метод конечного элемента. Определяющие уравнения конечного элемента справедливы в любой ортогональной плоской координатной системе или в криволинейной координатной системе. Исходя из общих уравнений элемента выводятся два специальных конечных элемента, все матрицы элементов которых задаются в явной форме.

1. Introduction

In recent years the finite element method has been applied with great success to problems in various areas, notable in structural analysis, fluid mechanics and heat transfer. More recently investigators [1] applied the finite element method to field equations in engineering and physics.

In the present paper now we develop a finite element formulation for the solution of the complex three dimensional SCHRÖDINGER wave equation. The equivalent finite element model yields a time dependent system of linear algebraic equations that can be solved for any time interval by the CRANK-NICHOLSON method. From the general element expressions a three dimensional isoparametric and a triangular finite element are developed and all relevant element matrices are explicitly given.

2. The Schrödinger wave Equation

In quantum mechanics the *time dependent SCHRÖDINGER equation* is usually written as [2]

$$H\left(\frac{h}{2\pi i} \frac{\partial}{\partial q}, q\right) \Psi(t, q) = i\hbar \frac{\partial}{\partial t} \Psi(t, q) \quad (1)$$

where h is PLANCK's constant, $\Psi(t, q)$ is the wave function and $H(p, q)$ denotes the HAMILTON function. For a particle of mass M the HAMILTON function is given by

$$H(p, q) = \frac{1}{2M} g_{ij} \dot{p}^i \dot{p}^j + E(q), \quad i, j = 1, 3, \quad (2)$$

and g_{ij} is the covariant metric tensor defined as

$$g_{ij} = \frac{\partial \mathbf{R}}{\partial q^i} \cdot \frac{\partial \mathbf{R}}{\partial q^j} \quad (3)$$

where $\mathbf{R}(q^i)$ denotes the space vector.

The wave function is a complex quantity, so that its real and imaginary parts can be considered as independent variables or, alternatively, Ψ and its conjugate $\bar{\Psi}$ can be considered as separate field variables, which can be varied independently. The product $\Psi\bar{\Psi}$ is to be real and physically represents the probability density at time t for the presence of the particle in the considered region.

A solution for the wave function and its conjugate is obtained if HAMILTON's functional is stationary. Thus,

$$\mathcal{L} = \int_{t_1}^{t_2} \int \int \int L \, d\mathcal{R} \, dt = \text{Stationary}. \quad (4)$$

The LAGRANGE density function L [2] extended for curvilinear coordinates is given by

$$L = -\frac{\hbar^2}{8\pi^2 M} g^{ij} \bar{\Psi}_{,i} \Psi_{,j} - \frac{\hbar}{4\pi i} (\bar{\Psi} \dot{\Psi} - \dot{\bar{\Psi}} \Psi) - \bar{\Psi} E \Psi, \quad (5)$$

where $E(q^1, q^2, q^3)$ is a potential field. The comma and dot denote the partial derivatives with respect to space and time, respectively. It should be noted that, if the same index appears twice, as a subscript and superscript, then the summation convention applies. g^{ij} is the covariant metric tensor defined by

$$g^{ij} g_{ij} = \delta_k^k \quad (6)$$

and δ_k^i is the KRONECKER symbol.

3. The Discretised Schrödinger Equation

The fundamental concept in the finite element method is the idealisation of a continuum $\mathcal{R}$ by an assembly of discrete elements interconnected at a finite number of 'nodal' points on the boundary of the element. For each element, admissible functions are assumed, and the application of a suitable variational principle yields a set of algebraic equations which have, in the present case, the wave function at the nodal points as unknowns of the problem.

We define now the *interpolation function* for $\Psi(x^1, x^2, x^3, t)$ over the region r of one element by [1]

$$\Psi(\mathbf{x}, t) = f_i(\mathbf{x}) \Psi^i(t), \quad i = 1, \text{ Number of nodes,}$$

similar

$$\bar{\Psi}(\mathbf{x}, t) = f_i(\mathbf{x}) \bar{\Psi}^i(t), \quad (7)$$

where $f(\mathbf{x})$ are suitable shape functions, Ψ^i and $\bar{\Psi}^i$ denote the nodal quantities. Substitution of equation (7) into the LAGRANGE function leads to

$$L = -\frac{h^2}{8\pi^2 M} g^{ij} f_{m,i} f_{n,j} \bar{\Psi}^m \Psi^n - \frac{h}{4\pi i} (f_m f_n \bar{\Psi}^m \dot{\Psi}^n - f_n f_m \Psi^n \dot{\bar{\Psi}}^m) - E f_m f_n \bar{\Psi}^m \Psi^n, \quad (8)$$

$$i, j = 1, 3, \quad m, n = 1, \quad \text{Number of nodes.}$$

The *discretised form of the often used stress-energy tensor* τ becomes

$$T_{ij} = f_{m,i} \frac{\partial L}{\partial (f_{n,j})} (\bar{\Psi}^m \bar{\Psi}^n + \Psi^m \Psi^n) - \delta_{ij} L,$$

$$T_{i4} = \frac{h}{4\pi i} f_{m,i} f_n (\bar{\Psi}^m \dot{\Psi}^n - \Psi^m \dot{\bar{\Psi}}^n),$$

$$m, n = 1, \quad \text{Number of nodes,} \quad i, j = 1, 3.$$

$$T_{4j} = f_m \frac{\partial L}{\partial (f_{n,i})} (\bar{\Psi}^m \bar{\Psi}^n + \dot{\Psi}^m \Psi^n),$$

$$T_{44} = \frac{h^2}{8\pi^2 M} g^{ij} f_{m,i} f_{n,j} \bar{\Psi}^m \Psi^n + E f_i f_j \bar{\Psi}^i \Psi^j,$$

In order for $\mathcal{L}$ to be a minimum the LAGRANGE function L is subject to the following requirements

$$(L, \dot{\Psi})' - L, \Psi = 0, \quad (L, \dot{\bar{\Psi}})' - L, \bar{\Psi} = 0. \quad (9)$$

Introducing equation (8) into (9) and (4) yields the *discretised form of the SCHRÖDINGER equation valid within the domain of one finite element* as

$$\frac{h^2}{8\pi^2 M} a_{mn} \Psi^n + \frac{h}{2\pi i} b_{mn} \dot{\Psi}^n + c_{mn} \Psi^n = 0, \quad (10)$$

$$-\frac{h^2}{8\pi^2 M} a_{mn} \bar{\Psi}^n + \frac{h}{2\pi i} b_{mn} \dot{\bar{\Psi}}^n - c_{mn} \bar{\Psi}^n = 0, \quad m, n = 1, \quad \text{Number of nodes per element,} \quad (11)$$

where

$$a_{mn} = \int_r g^{ij} f_{m,i}(\mathbf{x}) f_{n,j}(\mathbf{x}) dr, \quad (12)$$

$$b_{mn} = \int_r f_m(\mathbf{x}) f_n(\mathbf{x}) dr, \quad (13)$$

$$c_{mn} = \int_r E(\mathbf{x}) f_m(\mathbf{x}) f_n(\mathbf{x}) dr, \quad i, j = 1, 3, \quad m, n = 1, \quad \text{Number of nodes.} \quad (14)$$

The complex wave function can also be written in the form

$$\Psi(\mathbf{x}, t) = u(\mathbf{x}, t) + iv(\mathbf{x}, t) \quad (15)$$

and the interpolation functions can now be expressed as

$$u(\mathbf{x}, t) = f_i(\mathbf{x}) u^i(t), \quad v(\mathbf{x}, t) = f_i(\mathbf{x}) v^i(t); \quad (16)$$

hence

$$\Psi(\mathbf{x}, t) = f_i(\mathbf{x}) [u^i(t) + iv^i(t)], \quad \bar{\Psi}(\mathbf{x}, t) = f_i(\mathbf{x}) [u^i(t) - iv^i(t)], \quad (17)$$

where u^i and v^i are now taken as the nodal quantities of the finite element. Substituting these functions into equation (10) and (11) yields

$$\frac{h^2}{8\pi^2 M} a_{mn} (u^n + iv^n) + \frac{h}{2\pi i} b_{mn} (\dot{u}^n + i\dot{v}^n) + c_{mn} (u^n + iv^n) = 0, \quad (18)$$

$$\frac{h^2}{8\pi^2 M} a_{mn} (u^n - iv^n) - \frac{h}{2\pi i} b_{mn} (\dot{u}^n - i\dot{v}^n) + c_{mn} (u^n - iv^n) = 0. \quad (19)$$

The finite elements are assembled in a grid of N elements representing the considered problem. The assemblage of the element equations is straight forward and can be carried out in a number of ways [3]. For the sake of clarity in presentation, we briefly outline the *usual procedure of transformation from the local coordinates to the global coordinates*. Thus,

$$u^i = \sum_{j=1}^N J_{ij} U^j, \quad v^i = \sum_{j=1}^N J_{ij} V^j, \quad i = 1, \dots, M, \quad M = \text{Number of nodes per element}, \quad (20)$$

where J is called the *connection matrix* and U, V are the *vectors of the nodal quantities of all elements in the global system*. Hence, the *system matrices* become

$$\begin{aligned} A_{mn} &= \sum_i^M \sum_j^M J_{im} J_{jn} a_{ij}, & B_{mn} &= \sum_j^M \sum_i^M J_{im} J_{jn} b_{ij}, \\ C_{mn} &= \sum_i^M \sum_j^M J_{im} J_{jn} c_{ij}, & m, n &= 1, \dots, K, \quad K = \text{Number of nodes of the global system}. \end{aligned} \quad (21)$$

Clearly, the element matrices a_{ij} , b_{ij} and c_{ij} depend only on the specific finite element under consideration and the size of the rectangular matrix J_{im} depends partly on the number of nodes per element and partly on the overall size of the considered problem. After all elements are assembled the *system equations* become

$$\frac{h^2}{8\pi^2 M} A_{mn} (U^n + iV^n) + \frac{h}{2\pi i} B_{mn} (\dot{U}^n + i\dot{V}^n) + C_{mn} (U^n + iV^n) = 0, \quad (22)$$

$$\frac{h^2}{8\pi^2 M} A_{mn} (U^n - iV^n) - \frac{h}{2\pi i} B_{mn} (\dot{U}^n - i\dot{V}^n) + C_{mn} (U^n - iV^n) = 0. \quad (23)$$

By adding and subtracting above equations a more convenient system is obtained as

$$\frac{h^2}{4\pi^2 M} A_{mn} U^n + \frac{h}{\pi} B_{mn} \dot{V}^n + 2C_{mn} U^n = 0, \quad (24)$$

$$-\frac{h^2}{4\pi^2 M} A_{mn} V^n + \frac{h}{\pi} B_{mn} \dot{U}^n - 2C_{mn} V^n = 0. \quad (25)$$

The quantities $hB_{mn}U^n/\pi$ and $hB_{mn}V^n/\pi$ are the *generalised canonical momenta* at node m of an element and $h^2 A_{mn}/4M\pi^2 + 2C_{mn}$ represents the *discrete equivalent* of the HAMILTON function H for the particle while in the region $\mathcal{R}$.

We follow now the procedure of CRANK-NICHOLSON [4] and assume that the quantities $\dot{U}$ and $\dot{V}$ vary linearly with time over a time step Δt . Then the variation of U and V may be expressed as

$$U^n(t) = U^n(t - \Delta t) + \frac{\Delta t}{2} [\dot{U}^n(t - \Delta t) + \dot{U}^n(t)], \quad (26)$$

$$V^n(t) = V^n(t - \Delta t) + \frac{\Delta t}{2} [\dot{V}^n(t - \Delta t) + \dot{V}^n(t)], \quad (27)$$

where time (t) refers to present time and $(t - \Delta t)$ to preceding time. Writing equation (24) for present and preceding time and denoting the present time step by i and the preceding one by $(i - 1)$ leads to

$$\begin{aligned} \frac{h^2}{4\pi^2 M} A_{mn} U_i^n + \frac{h}{\pi} B_{mn} \dot{V}_i^n + 2C_{mn} U_i^n &= 0, \\ \frac{h^2}{4\pi^2 M} A_{mn} U_{i-1}^n + \frac{h}{\pi} B_{mn} \dot{V}_{i-1}^n + 2C_{mn} U_{i-1}^n &= 0. \end{aligned} \quad (28)$$

Adding these two equations and substitution in equation (27) yields an algorithm for the unknown present time nodal quantities as

$$G_{mn} U_i^n + \alpha B_{mn} V_i^n = -G_{mn} U_{i-1}^n + \alpha B_{mn} V_{i-1}^n, \quad (29)$$

where for clearer presentation G_{mn} and α are introduced as

$$G_{mn} = \frac{h^2}{4\pi^2 M} A_{mn} + 2C_{mn}, \quad \alpha = \frac{2h}{\pi \Delta t}$$

and similar equation (25) yields

$$-G_{mn} V_i^n + \alpha B_{mn} U_i^n = G_{mn} V_{i-1}^n + \alpha B_{mn} U_{i-1}^n, \quad (30)$$

$m, n = 1, 2, \dots, K$, K = Number of nodes of the global system. The expressions (29) and (30) are two coupled systems of $2 \times K$ linear equations representing the SCHRÖDINGER equation in discretised form for any geometrically flat or curved space-time continuum.

The solution of this set of equations is straight forward. The right hand sides of these equations are known quantities from the preceding time step. For the initial time $t = 0$, the nodal quantities $\{U\}_0$ and $\{V\}_0$ are the initial conditions of the time dependent portion of the wave function on the geometric boundary. The geometry and the boundary conditions are prescribed by the assembly of a specified number of finite elements which define the geometrical boundaries of a given problem in space. Hence, any convenient linear equation resolver may be applied to solve the above system for one time step. The obtained solution then is substituted as new right hand side and the system is solved again. The convergence of the described formulation is assured by the requirements of equation (9) together with certain requirements on the finite element shape functions as outlined in the next paragraph.

4. A three Dimensional Isoparametric Finite Element

As an example consider the compatible three dimensional finite element as shown in figure 1.

The critical step in the finite element procedure is the selection of the interpolation function for the main variable of the formulation. Selection of such functions for the variables u and v is constrained by certain admissibility requirements. Thus, for the functional L in equation (4) to be integrable it is necessary that u, v be continuous within the domain $\mathcal{R}$ of the integration. If the domain is broken into a number of subdomains r or finite elements then the satisfaction of the above condition requires also the continuity of u, v across the element boundaries.

In order for convergence to be rigorously assured the interelement compatibility must be enforced for the wave function and its derivatives up to the order $(n - 1)$, where n is the highest order derivative in the LAGRANGE density function (C^{n-1} -continuity). This implies that the admissible interpolation functions must be, at least, of order n . Higher order terms may be added to obtain the necessary number of parameter for a particular element configuration. From the expression (5) it is obvious that the highest order derivative is one and hence only C^0 -continuity has to be satisfied.

We now assume the *interpolation function* for the real and imaginary part of the wave function as follows

$$\begin{aligned} f_1 &= (1 - \xi)(1 - \eta)(1 - \zeta), & f_2 &= (1 - \xi)(1 + \eta)(1 - \zeta), & f_3 &= (1 + \xi)(1 + \eta)(1 - \zeta), \\ f_4 &= (1 + \xi)(1 - \eta)(1 - \zeta), & f_5 &= (1 - \xi)(1 - \eta)(1 + \zeta), & f_6 &= (1 - \xi)(1 + \eta)(1 + \zeta), \\ f_7 &= (1 + \xi)(1 + \eta)(1 + \zeta), & f_8 &= (1 + \xi)(1 - \eta)(1 + \zeta). \end{aligned} \quad (31)$$

For the given global coordinate system of figure 1 the finite element matrix (12) becomes

$$a_{mn} = \int_{(V_e)} \mathcal{L}^t f_m \mathcal{L} f_n \, dx \, dy \, dz, \quad m, n = 1, 2, \dots, 8, \quad (32)$$

where $\mathcal{L}$ is the linear operator

$$\mathcal{L}^t = \left[\frac{\partial}{\partial x} \frac{\partial}{\partial y} \frac{\partial}{\partial z} \right]. \quad (33)$$

The derivatives of the functions f_m in an isoparametric coordinate system are related to the derivatives in the CARTESIAN system as following

$$\mathcal{L}^t f_m = [J]^{-1} \mathcal{P} f_m, \quad (34)$$

where the operator $\mathcal{P}$ is defined as

$$\mathcal{P}^t = \left[\frac{\partial}{\partial \xi} \frac{\partial}{\partial \eta} \frac{\partial}{\partial \zeta} \right]. \quad (35)$$

$[J]$ denotes the JACOBIAN matrix

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix}. \quad (36)$$

The JACOBIAN matrix can be expressed in terms of the coordinates of the corner nodes x_i as follows

$$[J] = \frac{1}{8} \begin{bmatrix} -(1-\eta)(1-\zeta) & -(1+\eta)(1-\zeta) & (1+\eta)(1-\zeta) & (1-\eta)(1-\zeta) & -(1+\eta)(1+\zeta) & -(1-\eta)(1+\zeta) & (1+\eta)(1+\zeta) & (1-\eta)(1+\zeta) \\ -(1-\xi)(1-\zeta) & (1-\xi)(1-\zeta) & -(1+\xi)(1-\zeta) & -(1+\xi)(1-\zeta) & -(1-\xi)(1+\zeta) & -(1-\xi)(1+\zeta) & (1+\xi)(1+\zeta) & -(1+\xi)(1+\zeta) \\ -(1-\xi)(1-\eta) & -(1-\xi)(1+\eta) & -(1+\xi)(1+\eta) & -(1+\xi)(1-\eta) & (1-\xi)(1-\eta) & (1-\xi)(1+\eta) & (1+\xi)(1-\eta) & (1+\xi)(1+\eta) \end{bmatrix} \times$$
$$\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_8 & y_8 & z_8 \end{bmatrix} \quad (37)$$

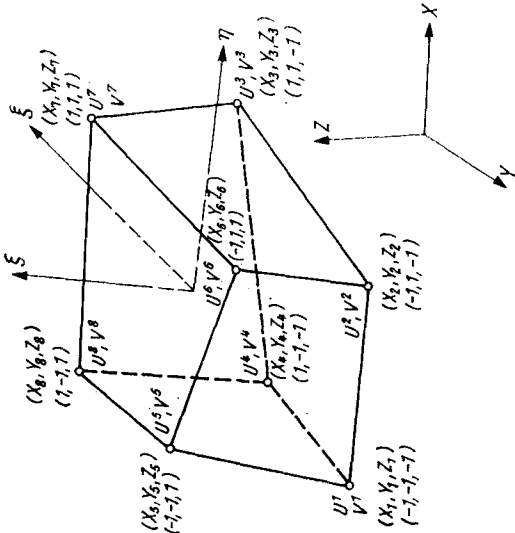


Fig. 1. Three dimensional Isoparametric Finite Element

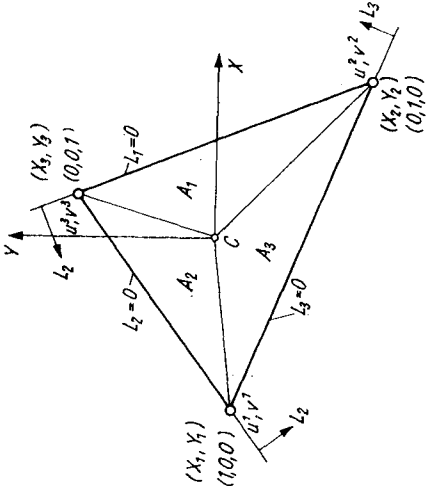


Fig. 2. Triangular Finite Element

Finally, the integral over the element domain $V_e(x, y, z)$ can be written as

$$\int_{(V_e)} f(\cdot) dV = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 f(\cdot) |\det [J]| d\xi d\eta d\zeta. \quad (38)$$

Substitution of above equations into the element matrices (12–14) yield the following governing matrices of this particular element as

$$a_{mn} = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \mathcal{P}^t f_m ([J]^{-1})^t [J]^{-1} \mathcal{P} f_n |\det [J]| d\xi d\eta d\zeta. \quad (39)$$

Evaluation of above expression, and setting $Q = J^{-1}$ for typographical convenience, leads to the final equation

$$\begin{aligned} a_{mn} = & \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 [(Q_{11}f_{m,\xi} + Q_{12}f_{m,\eta} + Q_{13}f_{m,\zeta})(Q_{11}f_{n,\xi} + Q_{12}f_{n,\eta} + Q_{13}f_{n,\zeta}) + \\ & + (Q_{21}f_{m,\xi} + Q_{22}f_{m,\eta} + Q_{23}f_{m,\zeta})(Q_{21}f_{n,\xi} + Q_{22}f_{n,\eta} + Q_{23}f_{n,\zeta}) + \\ & + (Q_{31}f_{m,\xi} + Q_{32}f_{m,\eta} + Q_{33}f_{m,\zeta})(Q_{31}f_{n,\xi} + Q_{32}f_{n,\eta} + Q_{33}f_{n,\zeta})] |\det [J]| d\xi d\eta d\zeta, \end{aligned} \quad (40)$$

$m, n = 1, \dots, 8,$

and similar

$$b_{mn} = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 f_m f_n |\det [J]| d\xi d\eta d\zeta, \quad (41)$$

$$c_{mn} = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 E(\xi, \eta, \zeta) f_m f_n |\det [J]| d\xi d\eta d\zeta. \quad m, n = 1, \dots, 8. \quad (42)$$

The integration of the latter expression can only be performed if the function of the potential field over the element domain is known. If for example the potential field over one element is approximated by a constant $E(\xi, \eta, \zeta) = E_0$, then

$$c_{mn} = E_0 b_{mn}.$$

The integration of the equations (40)–(42) is easily carried out by computer and the procedure used is usually the two point GAUSSIAN quadrature formula. The remaining work is the assemblage of the elements to the actual problem. This is straight forward and is also done by computer [3]. After assemblage the initial conditions will be applied on the surface nodes by specifying the values of the wave function U^n and V^n at these nodes. The introduction of the initial conditions make the global system nonsingular and the equations can be solved.

5. A two Dimensional Finite Element

Consider the triangular finite element as shown in figure 2. The triangle is defined by three points (x_i, y_i) with the origin of the coordinate system at the centroid. The shape functions are chosen to be polynomials of the area coordinates (L_1, L_2, L_3) [5]. The area coordinates are defined by $L_i = A_i/A$ and $L_1 + L_2 + L_3 = 1$ where A is the total area of an element. The differentiation of the CARTESIAN coordinates with respect to the area coordinates is given by

$$\frac{\partial}{\partial x} = \sum_{i=1}^3 \frac{b_i}{2A} \frac{\partial}{\partial L_i}, \quad \frac{\partial}{\partial y} = \sum_{i=1}^3 \frac{a_i}{2A} \frac{\partial}{\partial L_i}, \quad (43)$$

where

$$\begin{aligned} b_1 &= y_2 - y_3, & a_1 &= x_3 - x_2, \\ b_2 &= y_3 - y_1, & a_2 &= x_1 - x_3, \\ b_3 &= y_1 - y_2, & a_3 &= x_2 - x_1, \end{aligned} \quad (44)$$

and

$$2A = a_3 b_2 - a_2 b_3 = a_1 b_3 - a_3 b_1 = a_2 b_1 - a_1 b_2.$$

The integration of the element matrices can be performed by [5]

$$\int_{(A)} L_1^p L_2^q L_3^r dx dy dz = 2A \frac{p!q!r!}{(p+q+r+2)!}. \quad (45)$$

The *shape functions* for this element which satisfy C^0 -continuity are chosen to be

$$f_1(L_1, L_2, L_3) = L_1, \quad f_2(L_1, L_2, L_3) = L_2, \quad f_3(L_1, L_2, L_3) = L_3 \quad (46)$$

and the element matrices become

$$[a_{mn}] = \frac{1}{2A} \begin{bmatrix} (y_2 - y_3)^2 + (x_3 - x_2)^2 & \begin{array}{l} (y_2 - y_3)(y_3 - y_1) + \\ + (x_3 - x_2)(x_1 - x_3) \end{array} & \begin{array}{l} (y_2 - y_3)(y_1 - y_2) + \\ + (x_3 - x_2)(x_2 - x_1) \end{array} \\ \text{symmetrical} & (y_3 - y_1)^2 + (x_1 - x_3)^2 & \begin{array}{l} (y_3 - y_1)(y_1 - y_2) + \\ + (x_1 - x_3)(x_2 - x_1) \end{array} \\ & & (y_1 - y_2)^2 + (x_2 - x_1)^2 \end{bmatrix}, \quad (47)$$

$$b_{mn} = \frac{1}{4A} \left[B_{mn} + \frac{1}{12} b_m b_n \sum_{i=1}^3 x_i^2 + \frac{1}{12} a_m a_n \sum_{i=1}^3 y_i^2 + \frac{1}{12} (a_m b_n + a_n b_m) \sum_{i=1}^3 x_i y_i \right], \quad m, n = 1, 3, \quad (48)$$

where

$$[B_{mn}] = \begin{bmatrix} (x_2 y_3 - x_3 y_2)^2 & (x_2 y_3 - x_3 y_2)(x_3 y_1 - x_1 y_3) & (x_2 y_3 - x_3 y_2)(x_1 y_2 - x_2 y_1) \\ \text{symmetrical} & (x_3 y_1 - x_1 y_3)^2 & (x_3 y_1 - x_1 y_3)(x_1 y_2 - x_2 y_1) \\ & & (x_1 y_2 - x_2 y_1)^2 \end{bmatrix}; \quad (49)$$

$$c_{mn} = \left(\frac{1}{2A} \right)^2 \int_{(A)} E(x, y, z) f_m(x, y, z) f_n(x, y, z) dx dy, \quad (50)$$

where

$$\begin{aligned} f_1 &= (x_2 y_3 - x_3 y_2) + b_1 x + a_1 y, \\ f_2 &= (x_3 y_1 - x_1 y_3) + b_2 x + a_2 y, \\ f_3 &= (x_1 y_2 - x_2 y_1) + b_3 x + a_3 y. \end{aligned} \quad (51)$$

In the case where the potential field $E(x, y, z) = E_0$ can be assumed as constant over one element, the element matrix c_{mn} becomes

$$c_{mn} = E_0 b_{mn}.$$

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